GITHUB

DIFFERENCE BETWEEN GITHUB / GITLAB / BITBUCKET

GITHUB

1. it is owned by Microsoft
2. we can host repo on their servers
3. offers free and paid plans with varying features.
4. simple and initiative user interface.
5. offers a wide range of integrations with other tools and services.
6. robust security feature such as two factor authentication (2FA) code scanning and vulnerability alerts.
7. provide various third-party integration for continuous integration and continuous delivery work.
8. it having the most user-friendly interface.

GITLAB

1. it is independent company.
2. it provides both cloud-hosted and self-hosted version.
3. offers free and open-source community edition as well as paid plans with more advance feature and support.
4. more feature rich and customizable interface.
5. built in continuous integration and delivery (CI/CD) system.
6. offers built in container scanning and dependency scanning for added security.
7. offers its own pre-built continuous integration and continuous delivery support.
8. it has a steeper learning curve.

BITBUCKET

1. it is a cloud-based repo hosting services that is owned by Atlassian.
2. bitbucket offers free plans for small teams and paid plans for larger teams.
3. bitbucket has more built in tolls for project management and issue tracking.
4. bitbucket integrate seamlessly with other Atlassian tools such as JIRA and confluence.
5. it also offers security features.
6. it also having the most user-friendly interface.

GITTEA

1. It is a lightweight self-hosted Git service used to manage source code repositories.
2. It provides features like GitHub and GitLab such as repositories, branches, pull requests, and user management.
3. Gitea is fast, easy to install, and consumes very low system resources.
4. It is mainly used by developers and organizations to host private or public Git repositories on their own server.

GitLab CE (Community Edition)

1. Free & open-source → anyone can use without cost
2. Basic features → Git, CI/CD, issue tracking
3. Community support → no official support from GitLab
4. Best for students & small projects

GitLab EE (Enterprise Edition)

1. Paid version → requires subscription
2. Advanced features → security, compliance, monitoring
3. Official support from GitLab
4. Best for companies & large teams

GitHub IDE

1. The GitHub IDE is an online code editor powered by GitHub.
2. It lets you write, edit, and manage code directly in your browser without installing anything.

1.create a repository

2.edit the repo url
https://github.com/mayurpandit25/practice
(change .com ----> .dev)
https://github.dev/mayurpandit25/practice

3.To return to github
Change from .dev ----> .com

TEAMS COLLABORATION

1.create a specific repository

1. Click on repo settings --------> collaborators
2. add people -----> <add username/gmail> --------> add to repository
3. get invite notification on mail -----> accept invitation ----> now access the repo

2.How to assign specific access to a person of these repository

1. click on specific user
2. Click on Role / Permission option
3. Assign the required permission:

- Read → Only view
- Triage → Manage issues
- Write → Push code
- Maintain → Manage repo settings
- Admin → Full access

Note:

- Give Write access → for developers
- Give Read access → for viewers
- Give Admin access carefully
- Always use branch + pull request instead of direct push

INSTALL VS-CODE ON WINDOWS

HOSTING SELF-HOSTED GITTEA REPOSITORY ON UBUNTU

INSTALLATION OF GITTEA ON UBUNTU SERVER

1. INSTALL GIT

1. apt update
2. apt install git -y
3. git –version

2. CREATE A DEDICATED USER FOR GITEA

1. adduser --system --shell /bin/bash --gecos 'Git Version Control' --group --disabled-password --home /home/git git

3. CREATE REQUIRED DIRECTORIES

1. sudo mkdir -p /var/lib/gitea/{custom,data,log}
2. sudo chown -R git:git /var/lib/gitea/
3. sudo chmod -R 750 /var/lib/gitea/
4. sudo mkdir /etc/gitea
5. sudo chown root:git /etc/gitea
6. sudo chmod 770 /etc/gitea

4. INSTALL SQLITE (SIMPLE DATABASE)

1. sudo apt install sqlite3 -y

5. DOWNLOAD GITEA BINARY

1. wget -O gitea https://dl.gitea.com/gitea/1.24.0/gitea-1.24.0-linux-amd64
2. sudo mv gitea /usr/local/bin/
3. sudo chmod +x /usr/local/bin/gitea
4. gitea –version

6. CREATE SYSTEMD SERVICE

1. sudo vim /etc/systemd/system/gitea.service

[Unit]
Description=Gitea
After=network.target
[Service]
RestartSec=2s
Type=simple

User=git
Group=git
WorkingDirectory=/var/lib/gitea/
ExecStart=/usr/local/bin/gitea web --config /etc/gitea/app.ini
Restart=always
Environment=USER=git HOME=/home/git GITEA_WORK_DIR=/var/lib/gitea
[Install]
WantedBy=multi-user.target

7. START GITEA SERVICE

1. sudo systemctl daemon-reload
2. sudo systemctl enable gitea
3. sudo systemctl start gitea
4. sudo systemctl status gitea

8. SECURITY GROUP

1. allow port 3000/22/443/80 on sg

9. ACCESS GITEA WEB UI

1. http://server_ip:3000

10. MANAGE PERMISSIONS

1. change the database type to SQLITE3
2. Admin Account Settings

11. INSTALL GITEA OVER WEB

12. CREATE A REPOSITORY ON GITTEA

1. demo

13. ON SERVER

1. git config --global user.name "mayur"
2. git config --global user.email "mayur@example.com"
3. git clone http://13.126.68.131:3000/mayur/demo.git
4. cd demo
5. touch index.html
6. git add .
7. git commit -m "first commit"
8. git push

INSTALLATION OF GITTEA ON DOCKER

1. vim docker-compose.yaml

```
version: "3"
services:
  server:
    image: gitea/gitea:latest
    container_name: gitea
    restart: always
    ports:
      - "3000:3000"
      - "222:22"
    volumes:
      - ./gitea:/data
    environment:
      - USER_UID=1000
      - USER_GID=1000
```

2. docker-compose up -d
3. docker ps
4. http://public_ip:3000
